

AD 502 – Machine Learning

Pre-Reading Material – Lecture 21

Distribution Model Based Clustering

Why Do We Study Model-Based Clustering?

Model-based clustering is an **unsupervised learning approach** in which data is assumed to be generated from a mixture of underlying probability distributions.

Instead of simply grouping points based on distance, the method tries to identify the **statistical distribution that best represents each cluster**.

What is Distribution Model-Based Clustering?

In distribution model-based clustering, each cluster is represented by a **probability distribution**.

The algorithm assumes that the observed dataset is generated from a combination of several underlying distributions.

For example, a dataset may contain three groups of observations, where each group follows a different probability distribution.

Data → Probability Model → Identify Clusters

Basic Idea

The key idea is:

Each cluster represents a different probability distribution, and each data point has a probability of belonging to each cluster.

This is different from methods such as K-Means, where a data point is assigned to the nearest cluster center.

Model-based clustering can therefore provide a **soft assignment** of data points to clusters.

Gaussian Mixture Model (GMM)

A commonly used distribution-based clustering technique is the **Gaussian Mixture Model (GMM)**.

GMM assumes that the data is generated from a mixture of several **Gaussian (normal) distributions**.

Each Gaussian distribution can represent one cluster.

A cluster is characterized by properties such as:

- **Mean** – indicates the center of the distribution.
- **Variance/Covariance** – indicates the spread and shape of the distribution.
- **Weight** – represents the relative contribution of a cluster to the overall dataset.

Soft Clustering

One important feature of model-based clustering is **soft clustering**.

Instead of saying:

"This data point definitely belongs to Cluster 1."

the model can provide probabilities such as:

- Cluster 1 → high probability
- Cluster 2 → moderate probability
- Cluster 3 → low probability

The data point can then be associated with the cluster having the highest probability.

6. Expectation-Maximization (EM)

The **Expectation-Maximization (EM)** algorithm is commonly used to estimate the parameters of mixture models.

It works iteratively through two main ideas:

Expectation Step

Estimate how likely each data point is to belong to each cluster.

Maximization Step

Update the distribution parameters using these estimated memberships.

These steps are repeated until the model becomes stable or reaches a stopping condition.

Model-Based vs Partitioning Clustering

Feature	Partitioning Clustering	Model-Based Clustering
Basic idea	Groups data based on similarity/distance	Assumes an underlying probability model
Cluster representation	Often centroid	Probability distribution
Assignment	Usually hard assignment	Usually probabilistic/soft assignment
Example	K-Means	Gaussian Mixture Model
Cluster shape	K-Means works best with suitable distance-based structures	Can model different distribution shapes

Advantages

- Provides probabilistic cluster membership.
- Can model clusters with different shapes and spreads.
- Provides a statistical interpretation of clusters.
- Useful when the underlying data distribution is meaningful.

Limitations

- Requires assumptions about the underlying probability distribution.
- Results can depend on the selected model and initialization.
- Choosing the appropriate number of clusters can be challenging.
- Computational complexity can increase for large or high-dimensional datasets.

Learning Outcomes

After studying this topic, you should be able to:

1. Explain the concept of distribution model-based clustering.
2. Describe how probability distributions can represent clusters.
3. Explain the basic idea of Gaussian Mixture Models.
4. Differentiate between hard and soft cluster assignments.
5. Describe the basic role of the EM algorithm.

Think Before the Lecture

1. How is model-based clustering different from K-Means?
2. Why might a data point belong to more than one cluster with different probabilities?
3. What information does a Gaussian distribution provide about a cluster?
4. Why is the EM algorithm useful in mixture models?
5. What could happen if the assumed probability distribution does not match the actual data?

Quick Recall

Data → Assume Distribution Model → Estimate Cluster Membership → Update Model → Repeat → Final Clusters

Key idea: Distribution model-based clustering identifies groups by assuming that the data is generated from a mixture of underlying probability distributions.

